Lehrstuhl für Information System Engineering
TUM School of Computation, Information and Technology
Technische Universität München

Sustainable Software Service Engineering

Practical Course (IN0012) / Advanced Practical Course (IN2106)

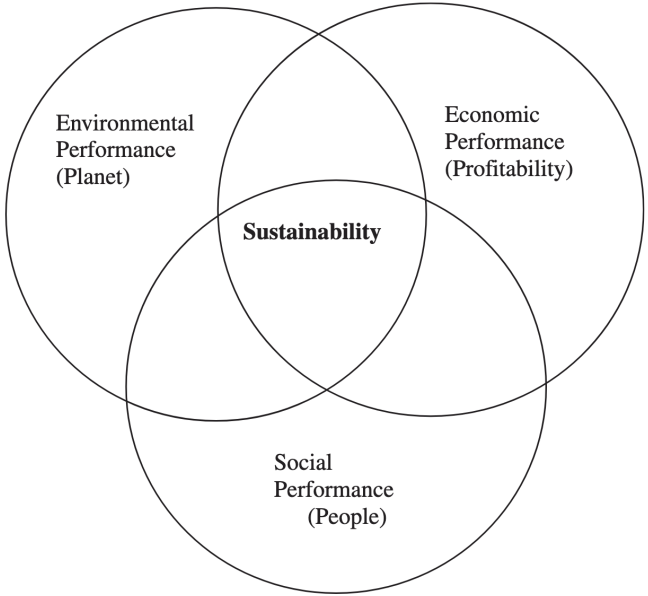
What is Sustainability?

"Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs."— World Commission on Environment and Development (WCED), 1987

Sustainability is commonly understood as the balance among *environmental, social, and economic* dimensions.

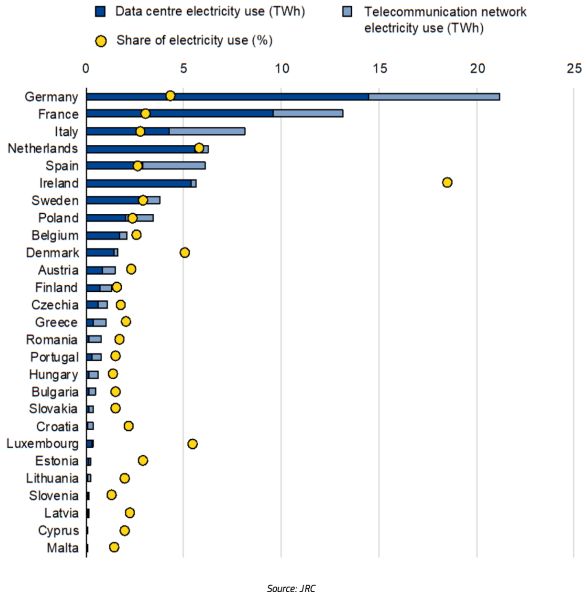
- Environmental / Planet: ecological integrity and resource conservation
- Social / People: equity, community, and human well-being
- Economic / Prosperity: viable economies and responsible growth

…increasingly complemented by *digital* sustainability: the practice of designing, operating, and applying digital technologies in ways that minimize harm and maximize long-term benefits for the environment, society, and the economy

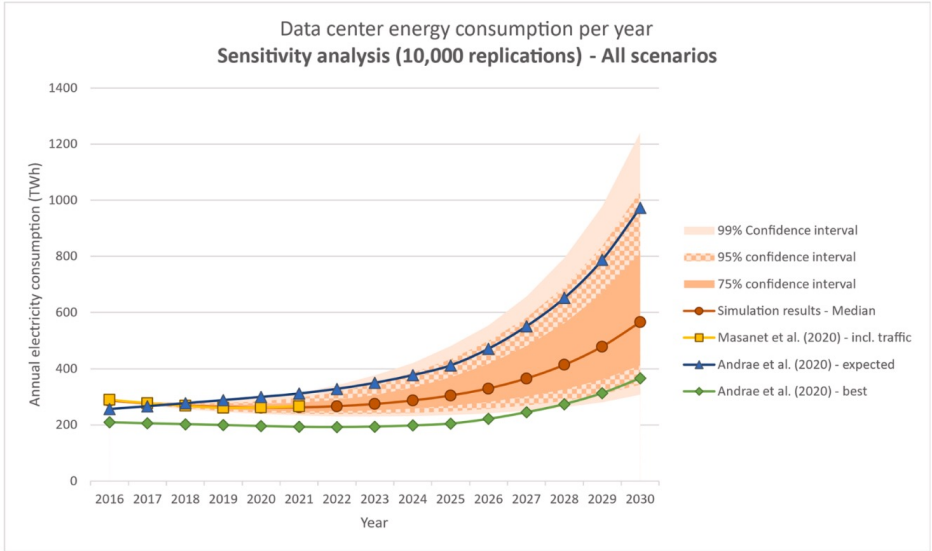


Dao, Viet et al. "From Green to Sustainability: Information Technology and an Integrated Sustainability Framework." (2011) https://doi.org/10.1016/j.jsis.2011.01.002.

Why It Matters



Kamiya, G. and Bertoldi, P., Energy Consumption in Data Centres and Broadband Communication Networks in **2022** the EU, 2024



M. Koot, F. Wijnhoven, “Usage impact on data center electricity needs: A system dynamic forecasting model” - **2021**

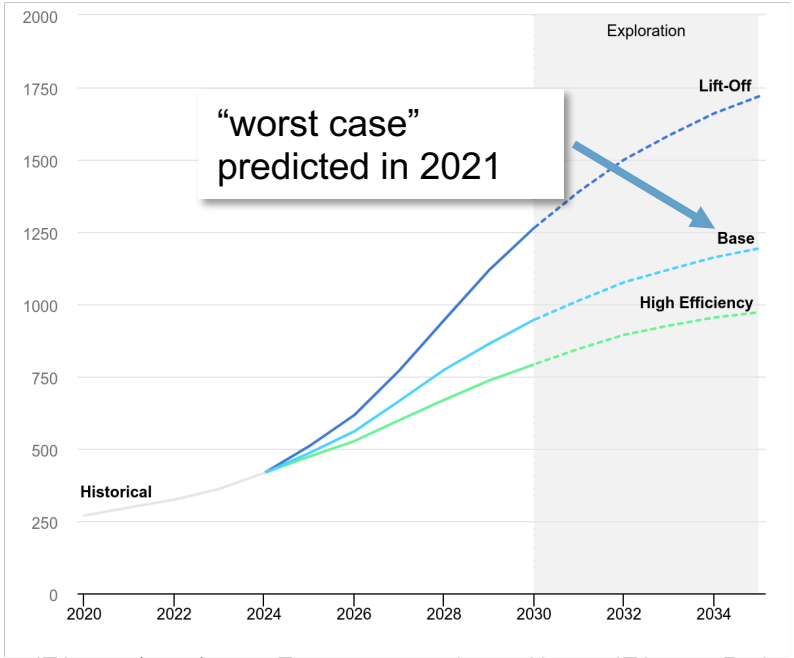
Why It Matters even more today

Lift-Off:

- Assumes stronger growth in AI adoption Global electricity demand is 45% higher
- Reaching 1700 TWh, which would be 4.4% of global electricity demand in 2035

High Efficiency:

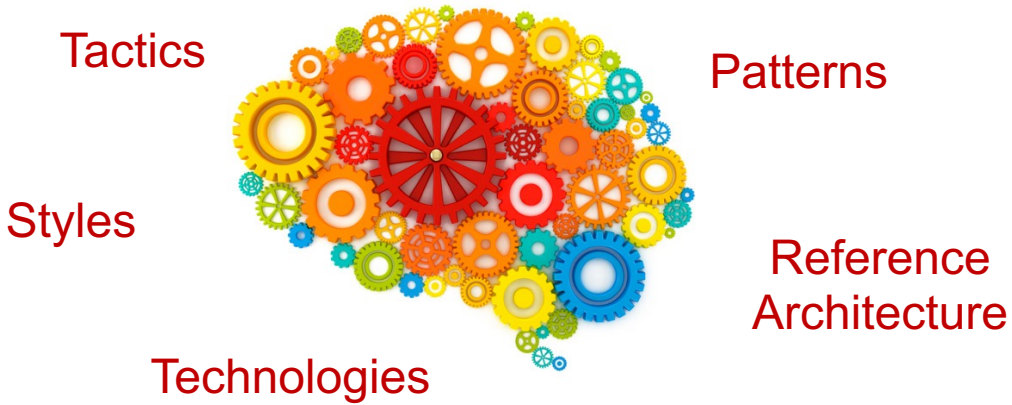
- Assumes stronger progress on energy efficiency in software, hardware, and infrastructure.
  That could be you!
- 15 % energy savings
- 970 TWh by 2035, resulting in 2.6% of the global electricity demand



IEA (2025), Energy and AI, IEA, Paris https://www.iea.org/reports/energy-and-ai, Licence: CC BY 4.0

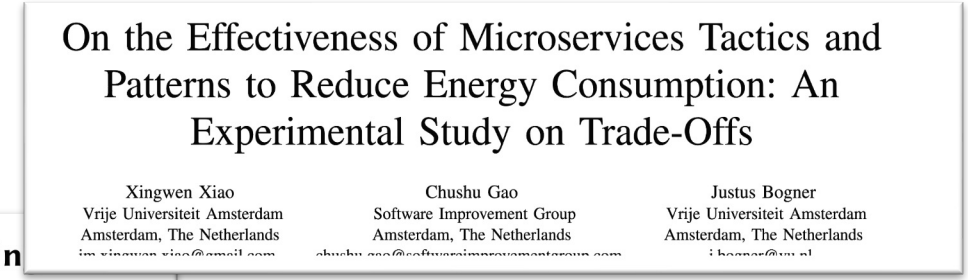
Architecture Solutions as Architecture Knowledge

„A good decision is based on knowledge“



... backed by real world evidence

Sustainable Engineering Tactics

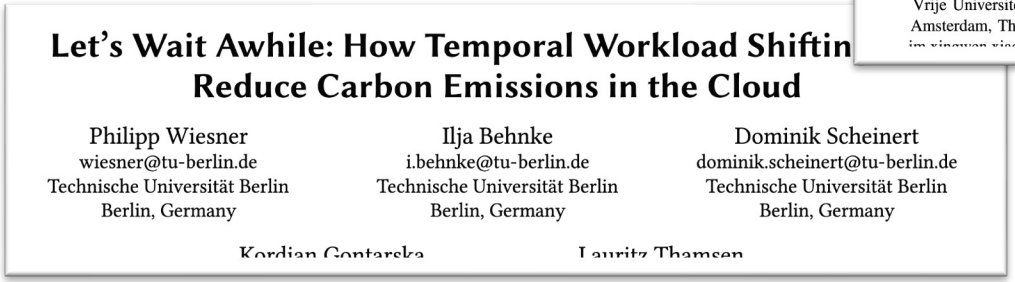


On the Effectiveness of Microservices Tactics and Patterns to Reduce Energy Consumption: An Experimental Study on Trade-Offs

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Amsterdam, The Netherlands

Chushu Gao
Software Improvement Group
Amsterdam, The Netherlands

Justus Bogner
Vrije Universiteit Amsterdam
Amsterdam, The Netherlands



Let's Wait Awhile: How Temporal Workload Shiftin
Reduce Carbon Emissions in the Cloud

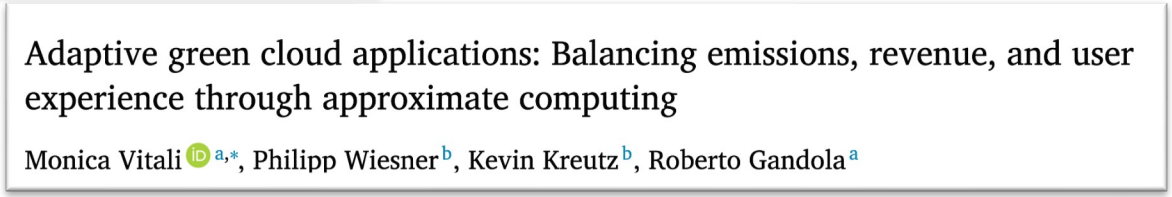
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Adaptive green cloud applications: Balancing emissions, revenue, and user experience through approximate computing

Monica Vitali [a,*], Philipp Wiesner [b], Kevin Kreutz [b], Roberto Gandola [a]

Sustainable Engineering Tactics

Bundling Requests, Caching Requests, Node-based balancing, Domain-Size

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Architecture Solutions as Architecture Knowledge

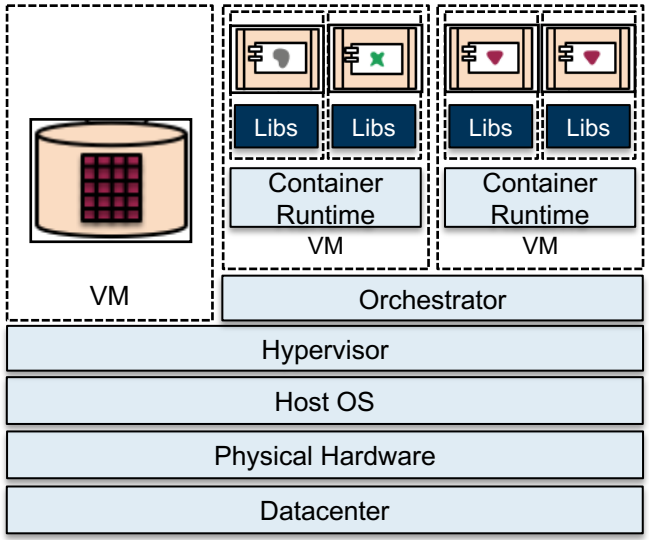
„A good decision is based on knowledge“



reproducible and verifiable measurements

~~... backed by real world evidence~~

What are Sustainable Measurements



Process Attribution

Host Level Attribution | RAPL | System Counters

Socket Measurements (IPMI, Redfin, PDU)

OS Level Attribution: Running Average Power Limit (RAPL)

CPUs typically implement the RAPL interface which allows us to get the power drawn by the CPU and other components of the system.

RAPL is composed of so called “Domains”

- **Core/PP0:** Energy consumed by the CPU Cores themselves.
- **Uncore/PP1**: Energy consumed by components close to the CPU (e.g. onboard GPU)
- **DRAM**: Energy consumed by the memory
- **Package/PKG**: "Core" and "Uncore“

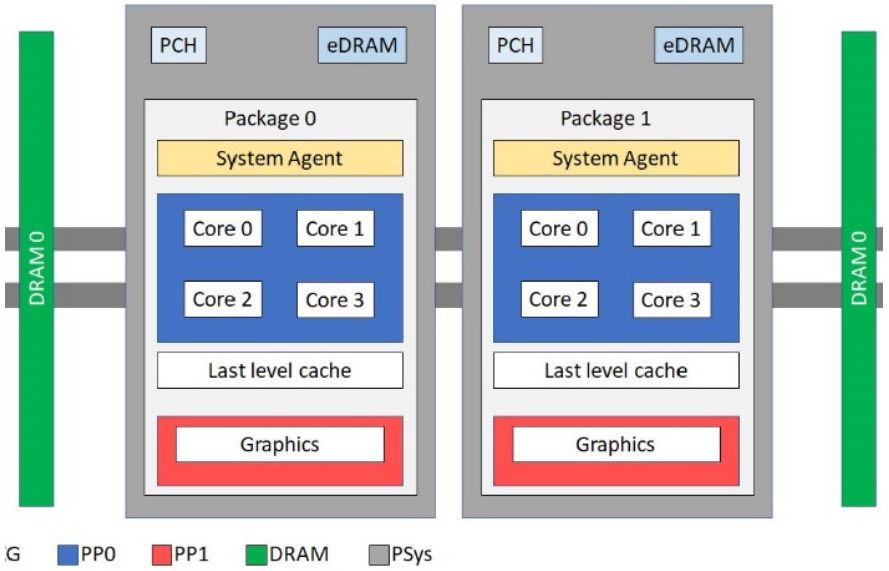


Figure 5. Power domains considered in RAPL interface.

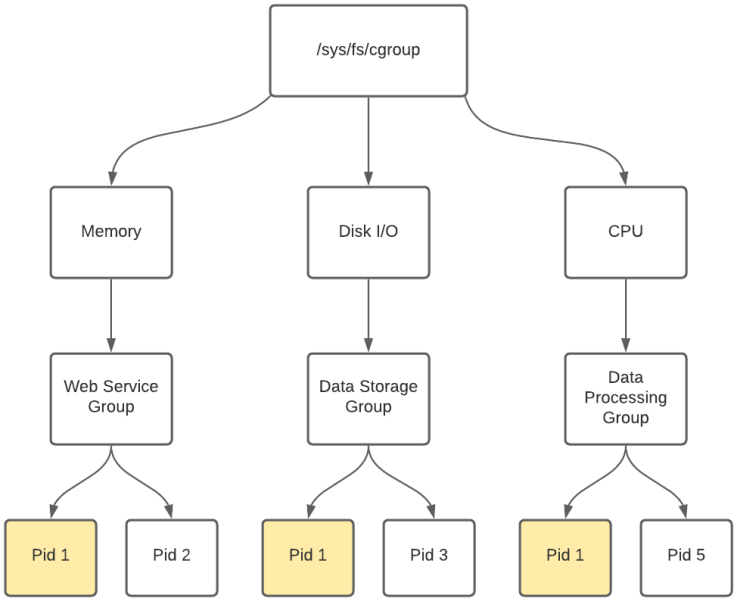
https://hubblo-org.github.io/scaphandre-documentation/explanations/rapl-domains.html

Process Level Attribution

Everything running on a machine is a process.

Processes can be grouped together in so called ***cgroups***. This is used for isolation mechanisms used by VM hypervisors.

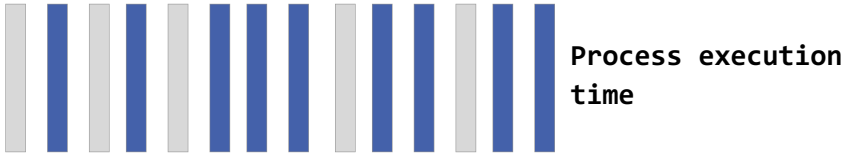
A nice side effect of this is, that we can easily determine which process is part of which VM and which processes are system processes.

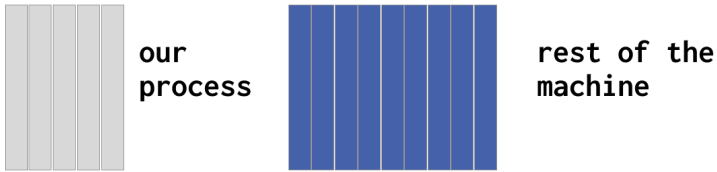


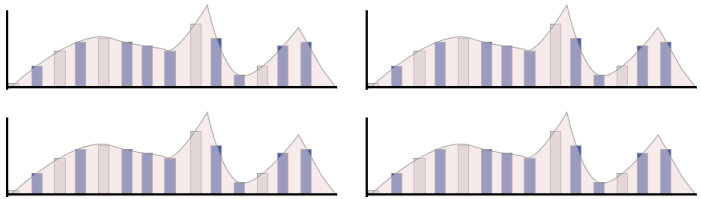
https://www.redhat.com/en/blog/cgroups-part-one

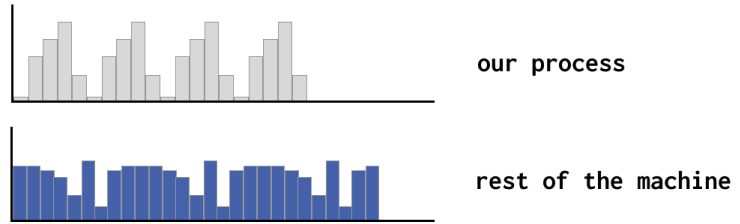
Process Level Attribution

Since everything running on a machine is a process. We can get the **process time**, the time a process spent in the CPU.



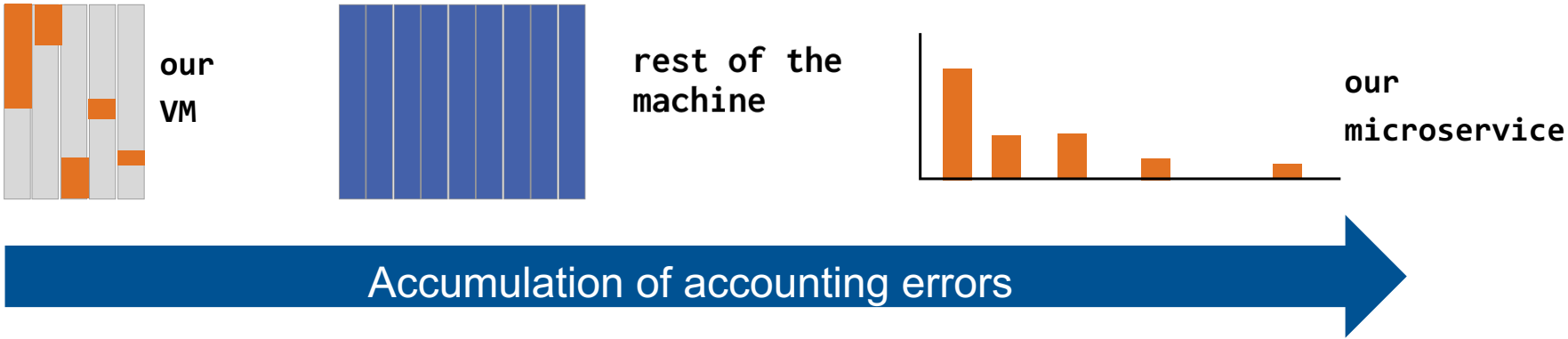




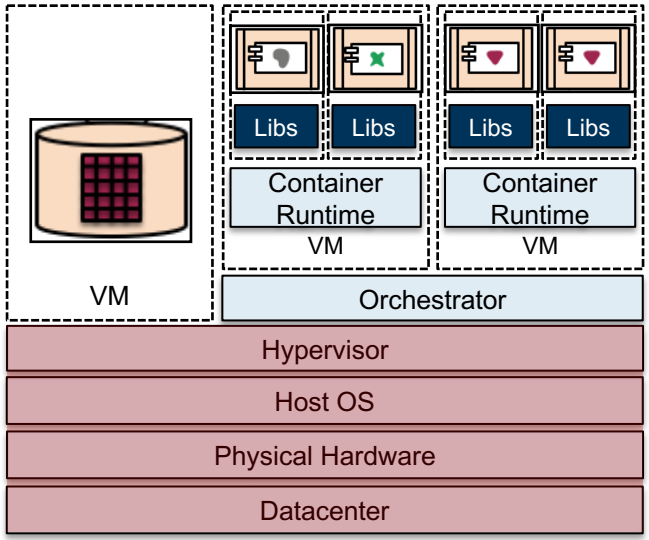


Pod Level Attribution

A VM is just another process from the host's perspective. So we need to identify which parts of the VM processes represent a pod or process we are interested in and then map them back to host processing time.



Problem in Cloud Environments



- Most real-world deployments run on ephemeral virtual hardware
- Access to bare-metal hardware/host
  - is expensive
  - Proprietary
- Access to RAPL can expose confidential data

How can we attribute the energy consumption of pods/microservices in virtualized environments without energy counters?

Your task if you chose to undertake this course

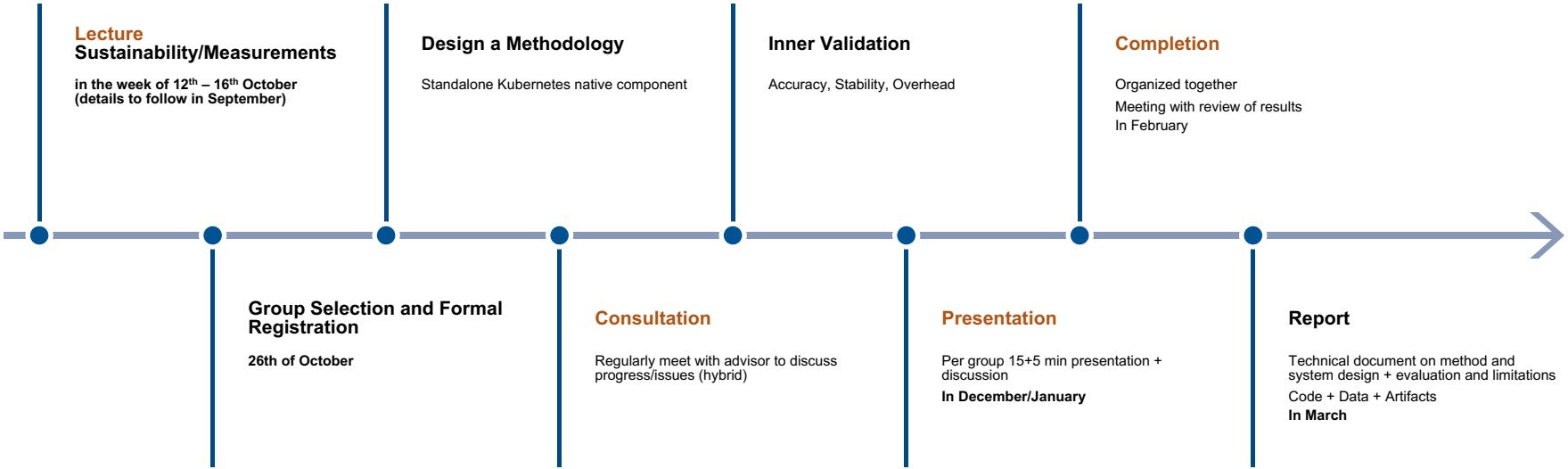
In groups of 2-3 students (mixed BA/MA)

1. Design a methodology for energy attribution in virtualized Kubernetes environments
   - Identify relevant algorithms and observable features for modeling the energy of pods (power model)
   - Prototype a method that fairly attributes power across all pods in a cluster
   - Implement an offline/online training approach
   - Design a validation strategy
   - Integrate it into Kubernetes/Prometheus

2. Identify 3+ test scenarios (applications) that run workloads in Kubernetes
   - Select 2 for testing/training
   - Nominate 1 that another team must validate against
   - Compare to wall power ground truth

3. We run a head-to-head comparison across all groups and workloads to see what methods work best by the end of the semester

Timeline


Lecture
Sustainability/Measurements
in the week of 12th – 16th October
(details to follow in September)
Group Selection and Formal Registration
26th of October
Design a Methodology
Standalone Kubernetes native component
Consultation
Regularly meet with advisor to discuss progress/issues (hybrid)
Inner Validation
Accuracy, Stability, Overhead
Presentation
Per group 15+5 min presentation + discussion
In December/January
Completion
Organized together
Meeting with review of results
In February
Report
Technical document on method and system design + evaluation and limitations
Code + Data + Artifacts
In March

Grading (tentative)

